

CSE310: Compiler Sessional, January 2026 Term

ICG Online Assignment Practice

Variable Initialization During Declaration

Time: 40 Mins

Total Marks: 10

Problem

In the provided grammar, variables are declared but initialized in a separate statement. You are required to support immediate initialization during declaration. Thus, your program should support the following syntax.

```
int a = 10, b = 20 + a;
```

Task

- Modify the `declaration_list` in the grammar to optionally accept an assignment `ID = expression`.
- Update the visitor to evaluate the expression and store the result immediately in the data segment (for global scopes) or in the computed stack offset (for local functions).

Mark Distribution

- Extending the grammar for initialization during declaration. (3)
- Generating code for global variable initialization. (3)
- Generating code for local variable initialization. (4)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)
<pre>int a = 5; int main() { int b = a * 2; println(a); println(b); return 0; }</pre>	<pre>segment readable writeable a dd 1 DUP (5) segment readable executable main: PUSH EBP MOV EBP, ESP MOV EAX, [a] MOV ECX, EAX MOV EAX, 2 IMUL ECX SUB ESP, 4 MOV [EBP-4], EAX MOV EAX, [a] CALL print_number MOV EAX, [EBP-4] CALL print_number MOV EAX, 0 JMP .L1 .L1: ADD ESP, 4 POP EBP RET</pre>

Expected Output	5 10
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